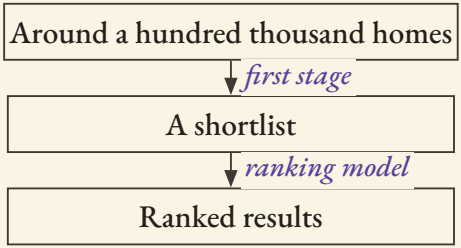
Airbnb: a shortlist out of a hundred thousand homes

Airbnb is an online marketplace for short stays in private homes.

The first stage of its search evaluates tens of thousands of homes and passes a shortlist to the ranking model.



Abdool and others (Airbnb), preprint arXiv, January 2026; Airbnb's engineering blog, April 2025.

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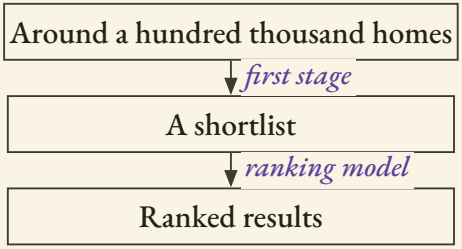
That stage used to be a simple linear model that scored every home once a day, whatever the search.



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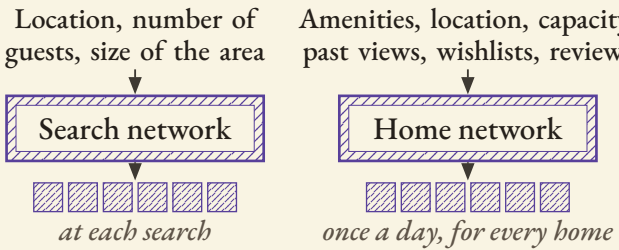
Two networks, and the homes nearest to the search

Searches and homes become lists of numbers.



Abdool and others (Airbnb), preprint arXiv, January 2026; Airbnb's engineering blog, April 2025.

Two networks, and the homes nearest to the search

Searches and homes become lists of numbers.

Location, number of guests, size of the area

Search network

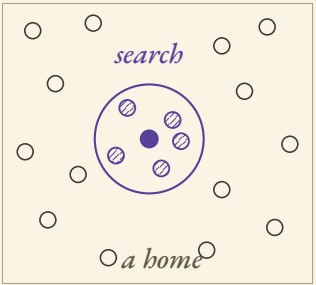
at each search

Amenities, location, capacity; past views, wishlists, reviews

Home network

once a day, for every home

The homes nearest to the search join the shortlist



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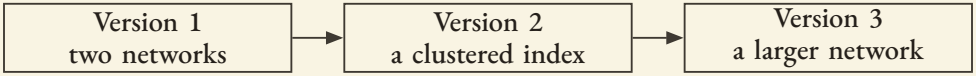
Three A/B tests, together +0.31% conversion



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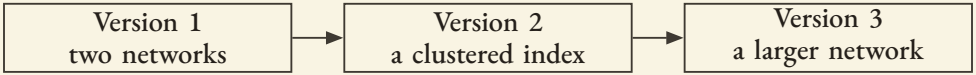

+0.31%

conversion, the three versions together, relative change

Airbnb's paper: "a large portion of the team's annual goal".

Airbnb's blog: on par with some of its largest machine learning gains in search ranking of the past two years.

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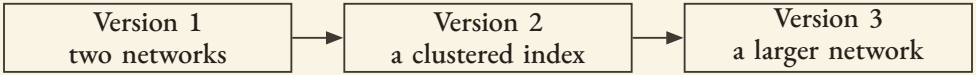

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+2.3% *bookings from promotional emails*

−16% *compute: the clustered index against the old first stage*

Abdool and others (Airbnb), preprint arXiv, January 2026; Airbnb's engineering blog, April 2025.